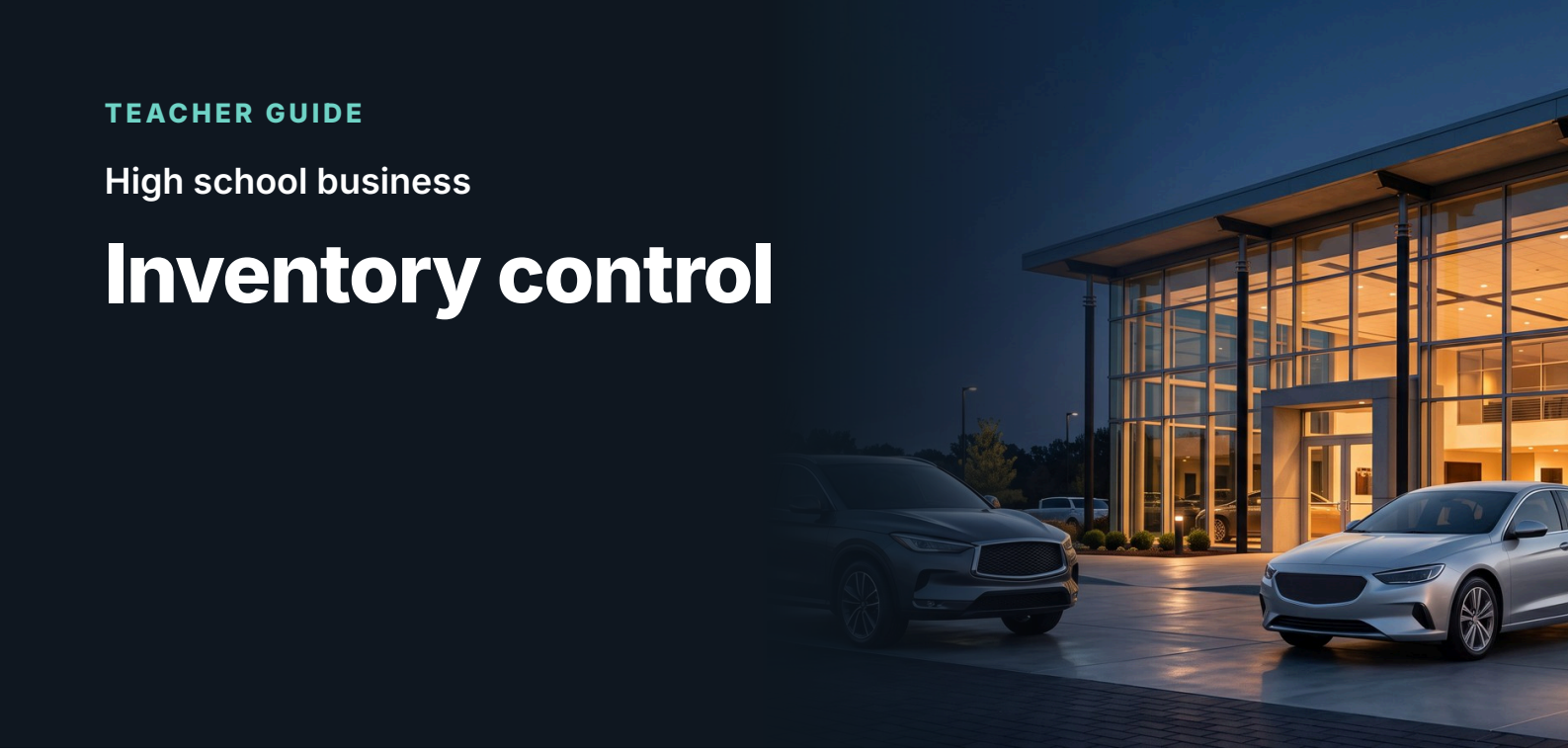
TEACHER GUIDE

High school business

Inventory control



A review lesson. Students run a car dealership in firms of four and review inventory control on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min | **45 min** | 60 min | 75 min

What students do

Every student buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal inventory

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Reorder point

Round 1

Never run out for them.

Reorder point, in cars?

They buy each week	**2 cars**
Supplier delivers in	**2 weeks**
Safety stock	**5 cars**

ANSWER

9 cars

2 × 2 + 5 = 9 cars

Why: The safety stock of 5 covers a late delivery or a busy week.

2 Too much inventory

Round 1

Your lot holds far more cars than you sell in a month.

Which two problems are most likely?

- **A** Lower insurance costs
- **B** Running out of cars to sell
- **C** Higher storage costs
- **D** Cash tied up in inventory
- **E** Lost sales from empty space

ANSWER

✓ Higher storage costs
✓ Cash tied up in inventory

Why: Extra inventory holds cash and costs money to keep.

3 Reorder in order

Round 2

You use a reorder point and safety stock.

Put these in order, first to last?

- **A** You place an order
- **B** New cars arrive
- **C** Inventory falls to the reorder point
- **D** The lead time passes

ANSWER

1. Inventory falls to the reorder point
2. You place an order
3. The lead time passes
4. New cars arrive

Why: The order goes in at the reorder point and arrives after the lead time.

Topic tasks

4 Cars to order now

Round 2

Fill the lot on delivery.

Cars to order now, to fill the lot on delivery?

In inventory now	**13 cars**
They buy each week	**5 cars**
Supplier delivers in	**2 weeks**
Your lot holds at most	**15 cars**

ANSWER

12 cars

13 − 5 × 2 = 3 left; 15 − 3 = 12 cars

Why: 3 are left when the order lands, so order 12.

5 JIT or hold cars

Round 3

Your supplier is two hours away and always on time. Hold 40 cars, or hold 10 and reorder often?

Which should you choose, and why?

A Hold 10 and reorder often

B Hold 40 cars

ANSWER

✔ **Hold 10 and reorder often**

Why: A reliable, nearby supplier lets you hold less and free up cash.

6 Weeks until reorder

Round 3

When do you order next?

Weeks until you reorder?

They buy each week	**3 cars**
Supplier delivers in	**3 weeks**
Safety stock	**2 cars**
In inventory	**14 cars**

ANSWER

1 week

Reorder point 3 × 3 + 2 = 11; (14 − 11) ÷ 3 = 1

Why: Reorder in 1 weeks, when 11 cars are left.

Final questions

Each student answers 6 final questions. They come from these 10 questions on inventory.

1. Your lot has 10 spaces and holds 6 vehicles. How many more vehicles can you buy before you need more space?
   ✔ **4**

2. You can buy 4 extra lot spaces for $6,000. What is that per space?
   ✔ **$1,500**

3. You buy 5 EV city cars listed at $9,000 each; five or more get 6% off. What does the order cost?
   ✔ **$42,300**

4. Does your firm pay sales tax when it buys cars from Wholesale to sell on?
   ✔ **No: a purchase for resale is exempt**

5. What is "inventory at cost" in your net position?
   ✔ **What the firm paid for the vehicles it still holds**

6. You hold 3 Panel vans bought at $15,000 each and sell one to a customer for $18,000. What happens before the return is filed?
   ✔ **Inventory at cost falls to $30,000; cash rises by $19,260**

7. You hold 4 unsold cars that cost $16,000 each. Sold to customers at $19,000 each, how much cash would they bring in, before filing the return?
   ✔ **$81,320**

8. Why buy cars mainly when there is a customer or a deal for them?
   ✔ **Unsold cars tie up cash and lot spaces**

9. Three or four cars get 4% off. You buy 4 Forklifts listed at $12,000 each. How much do you save against the list price?
   ✔ **$1,920**

10. You have $30,000 cash and 2 free spaces. Family SUVs list at $16,000; two cars get 2% off. How many can you buy, and what cash is left?
    ✔ **One, leaving $14,000**